

Walmart Sales Data Analysis using SQL

1. BASIC DATA UNDERSTANDING

Objective: Validate dataset scale and structure.

1.1 Total number of transactions

Business Question:

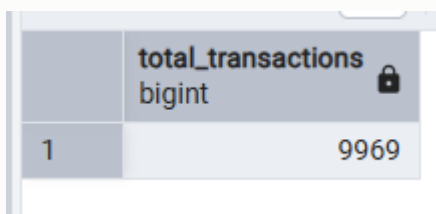
How many total sales transactions are available?

Purpose:

Measures dataset size and validates transaction volume.

```
SELECT COUNT(*) AS total_transactions FROM walmart;
```

Output:



	total_transactions bigint
1	9969

Insight:

The dataset contains **~10K transactions**, providing sufficient volume for branch, category, and time-based analysis.

Business Implication:

The dataset is large enough to support reliable branch-level, category-level, and time-based business decisions.

1.2 View first 5 rows of data

Business Question:

What does a typical transaction record look like?

Purpose:

Sanity check for schema, column values, and data correctness.

```
SELECT * FROM walmart LIMIT 5;
```

Output:

invoice_id bigint	branch text	city text	category text	unit_price double precision	quantity double precision	date timestamp without time zone	time time without time zone	payment_method text	rating double precision	profit_margin double precision	total double precision	hour integer	day integer	month integer	weekday text	is_weekend boolean	shift text	year integer	
1	1	WALMD...	San Anto...	Health and beauty	74.69	7	2019-05-01 00:00:00	13:08:00	Ewallet	9.1	0.48	322.82999999999999	13	1	5	Wednesd...	false	Afterno...	2019
2	2	WALMD...	Hartingen	Electronic accessori...	15.28	5	2019-08-03 00:00:00	10:29:00	Cash	9.6	0.48	76.39999999999999	10	3	8	Saturday	true	Morning	2019
3	3	WALMD...	Haltom Cl...	Home and lifestyle	46.33	7	2019-03-03 00:00:00	13:23:00	Credit card	7.4	0.33	324.31	13	3	3	Sunday	true	Afterno...	2019
4	4	WALMD...	Bedford	Health and beauty	58.22	8	2019-01-27 00:00:00	20:33:00	Ewallet	8.4	0.33	465.76	20	27	1	Sunday	true	Evening	2019
5	5	WALMD...	Irving	Sports and travel	86.31	7	2019-08-02 00:00:00	10:37:00	Ewallet	5.3	0.48	604.17000000000001	10	2	8	Friday	false	Morning	2019

Insight:

Each transaction captures **branch, city, category, pricing, quantity, payment method, rating, profit margin, and time features.**

Business Implication:

Confirms the dataset contains all required transactional attributes for multi-dimensional business analysis.

1.3 Unique branches, cities, and categories

Business Question:

How diverse is Walmart's operational coverage?

Purpose:

Evaluates geographic reach and product category spread.

```
SELECT COUNT(DISTINCT branch) AS total_branches,  
       COUNT(DISTINCT city) AS total_cities,  
       COUNT(DISTINCT category) AS total_categories  
FROM walmart;
```

Output:

	total_branches bigint	total_cities bigint	total_categories bigint
1	100	98	6

Insight:

Walmart operates across a **wide geographic footprint**, but with a **limited set of core product categories**.

Business Implication:

Walmart's wide geographic presence with a limited category mix enables localized optimization without expanding product complexity.

1.4 Maximum and minimum quantity sold

Business Question:

What is the range of items purchased per transaction?

Purpose:

Identifies buying behavior and detects extreme purchase patterns.

```
SELECT MAX(quantity) AS max_quantity, MIN(quantity) AS  
min_quantity FROM walmart;
```

Output:

	max_quantity double precision	min_quantity double precision
1	10	1

Insight:

Customers typically purchase in **small quantities**, with no extreme bulk-buying behavior.

Business Implication:

Customer purchases are predominantly low-quantity, supporting high-frequency retail and inventory replenishment strategies.

2. CUSTOMER & PAYMENT INSIGHTS

Objective: Understand customer payment behavior.

2.1 Transactions and quantity by payment method

Business Question:

Which payment methods are most used by customers?

Purpose:

Reveals customer payment preferences and transaction distribution.

```
SELECT
    payment_method,
    COUNT(*) AS num_transactions,
    SUM(quantity) AS total_quantity_sold
FROM walmart
GROUP BY payment_method
ORDER BY total_quantity_sold DESC;
```

Output:

	payment_method text	num_transactions bigint	total_quantity_sold double precision
1	Credit card	4256	9567
2	Ewallet	3881	8932
3	Cash	1832	4984

Insight:

Digital payments dominate, with credit cards and e-wallets accounting for the majority of transactions and quantity sold.

Business Implication:

Walmart should prioritize digital payment infrastructure to improve checkout efficiency and reduce cash-handling overhead.

2.2 Most preferred payment method per branch

Business Question:

Which payment method dominates at each branch?

Purpose:

Supports branch-level checkout and payment infrastructure planning.

```
WITH payment_ranking AS (  
    SELECT  
        branch,  
        payment_method,  
        COUNT(*) AS transaction_count,  
        RANK() OVER (PARTITION BY branch ORDER BY COUNT(*) DESC)  
AS rank  
    FROM walmart  
    GROUP BY branch, payment_method  
)  
SELECT * FROM payment_ranking WHERE rank = 1;
```

Output:

	branch text	payment_method text	transaction_count bigint	rank bigint
1	WALMO...	Ewallet	45	1
2	WALMO...	Ewallet	37	1
3	WALMO...	Credit card	115	1
4	WALMO...	Ewallet	44	1
5	WALMO...	Ewallet	56	1
6	WALMO...	Ewallet	50	1
7	WALMO...	Ewallet	52	1
8	WALMO...	Ewallet	39	1
9	WALMO...	Credit card	139	1
10	WALMO...	Ewallet	47	1
11	WALMO...	Ewallet	39	1
12	WALMO...	Ewallet	52	1
13	WALMO...	Ewallet	44	1
14	WALMO...	Ewallet	28	1
15	WALMO...	Ewallet	57	1
16	WALMO...	Ewallet	46	1

Insight:

Most branches show a **single dominant payment method**, with **e-wallets** appearing most frequently as the top choice.

Business Implication:

Branch-specific payment preferences indicate the need for localized payment strategies rather than a uniform national approach.

3. PRODUCT CATEGORY & PROFIT ANALYSIS

Objective: Identify revenue and profitability drivers.

3.1 Revenue and profit by category

Business Question:
Which product categories drive revenue and profit?

Purpose:
Compares category-level performance and profitability contribution.

```
SELECT
    category,
    SUM(total) AS total_revenue,
    SUM(total * profit_margin) AS total_profit
FROM walmart
GROUP BY category
ORDER BY total_profit DESC;
```

Output:

	category text	total_revenue double precision	total_profit double precision
1	Fashion accessories	489480.89999999997	192314.89320000037
2	Home and lifestyle	489250.06	192213.63809999999
3	Electronic accessori...	78175.02999999998	30772.489499999978
4	Food and beverages	53471.280000000006	21552.862200000003
5	Sports and travel	52497.930000000002	20613.808199999996
6	Health and beauty	46851.179999999998	18671.7345

Insight:

Revenue is **highly concentrated in two categories**, while others contribute marginally to overall profit.

Business Implication:

Business focus should be directed toward high-revenue and high-profit categories to maximize return on inventory and promotions.

3.2 Rating analysis by city and category

Business Question:

How does customer satisfaction vary across locations and categories?

Purpose:

Identifies high- and low-performing city–category combinations.

```
SELECT
    city,
    category,
    AVG(rating) AS avg_rating,
    MIN(rating) AS min_rating,
    MAX(rating) AS max_rating
FROM walmart
GROUP BY city, category
ORDER BY avg_rating DESC;
```

Output:

	city text	category text	avg_rating double precision	min_rating double precision	max_rating double precision
1	College Station	Health and beauty	10	10	10
2	DeSoto	Health and beauty	9.9	9.9	9.9
3	Rosenberg	Health and beauty	9.9	9.9	9.9
4	Mineral Wells	Health and beauty	9.8	9.8	9.8

Insight:

Customer satisfaction is **location-dependent**, even within the same category.

Business Implication:

Customer satisfaction varies by location, suggesting opportunities to replicate best practices across underperforming cities.

3.3 Highest-rated product category per branch

Business Question:

What category performs best at each branch?

Purpose:

Supports branch-specific product and merchandising decisions.

```
WITH category_rating AS (  
    SELECT  
        branch,  
        category,  
        AVG(rating) AS avg_rating,  
        RANK() OVER (PARTITION BY branch ORDER BY AVG(rating)  
DESC) AS rank  
    FROM walmart  
    GROUP BY branch, category  
)  
SELECT * FROM category_rating WHERE rank = 1;
```

Output:

	branch text	category text	avg_rating double precision	rank bigint
1	WALM0...	Electronic accessori...	7.45	1
2	WALM0...	Food and beverages	8.25	1
3	WALM0...	Sports and travel	7.5	1
4	WALM0...	Food and beverages	9.3	1

Insight:

Each branch has **a different best-performing category** in terms of customer satisfaction.

Business Implication:

Branch-level merchandising and promotions should emphasize locally preferred categories to improve customer experience.

3.4 Category share in total revenue

Business Question:

How much does each category contribute to overall revenue?

Purpose:

Highlights revenue concentration and category dependency.

```
SELECT
    category,
    ROUND((SUM(total) * 100.0 / (SELECT SUM(total) FROM
walmart))::numeric, 2) AS category_percent
FROM walmart
GROUP BY category
ORDER BY category_percent DESC;
```

Output:

	category text	category_percent numeric
1	Fashion accessories	40.46
2	Home and lifestyle	40.44
3	Electronic accessori...	6.46
4	Food and beverages	4.42
5	Sports and travel	4.34
6	Health and beauty	3.87

Insight:

Nearly **80% of revenue comes from just two categories**, indicating high revenue dependency.

Business Implication:

Revenue concentration in a few categories introduces risk, making diversification and performance improvement in weaker categories important.

4. TIME-BASED SALES TRENDS

Objective: Analyze demand patterns over time.

4.1 Sales by day of the week

Business Question:

Which days generate the highest sales?

Purpose:

Supports staffing, promotions, and scheduling decisions.

```
SELECT
    weekday,
    COUNT(*) AS num_transactions,
    SUM(total) AS revenue
FROM walmart
GROUP BY weekday
ORDER BY revenue DESC;
```

Output:

	weekday text	num_transactions bigint	revenue double precision
1	Tuesday	1479	184200.78000000003
2	Sunday	1466	182409.70999999996
3	Saturday	1411	176043.18
4	Wednesd...	1423	171501.57999999996
5	Thursday	1426	170830.39
6	Friday	1405	169546.83
7	Monday	1359	155193.91000000003

Insight:

Sales and transactions peak on **Tuesday, Sunday, and Saturday**, while **Monday** shows the lowest activity.

Business Implication:

Staffing and marketing efforts should be aligned with high-traffic weekdays to maximize operational efficiency.

4.2 Sales by hour

Business Question:

What are the peak sales hours?

Purpose:

Identifies high-demand periods for operational optimization

```
SELECT
    hour,
    COUNT(*) AS num_transactions,
    SUM(total) AS revenue
FROM walmart
GROUP BY hour
ORDER BY hour;
```

Output:

	hour integer	num_transactions bigint	revenue double precision
1	6	311	30839
2	7	338	35325
3	8	299	29591
4	9	325	34295
5	10	411	60756.22
6	11	403	60700.790000000001
7	12	409	57820.649999999994
8	13	436	66288.74
9	14	400	61472.3800000000005
10	15	1191	142016.770000000002
11	16	1173	134918.07
12	17	1027	116301.16
13	18	986	113072.8
14	19	1024	128581.06
15	20	972	109066.74
16	21	135	15182
17	22	126	13236
18	23	3	263

Insight:

Sales rise sharply from **11 AM onward**, peaking between **12 PM – 7 PM**, and drop significantly after **9 PM**.

Business Implication:

Peak sales hours require increased staffing and inventory availability, while low-demand hours can be optimized for cost savings.

4.3 Sales by shift

Business Question:

Which business shift is most active?

Purpose:

Helps optimize shift-wise staffing and resources.

```
SELECT
    branch,
    shift,
    COUNT(*) AS num_transactions
FROM walmart
GROUP BY branch, shift
ORDER BY branch, num_transactions DESC;
```

Output:

	branch text	shift text	num_transactions bigint
1	WALM0...	Afterno...	36
2	WALM0...	Evening	30
3	WALM0...	Morning	8

Insight:

Afternoon and Evening shifts dominate transaction volume, while Morning contributes the least.

Business Implication:

Resource allocation should favor Afternoon and Evening shifts to match higher customer demand.

4.4 Month-over-month revenue trend

Business Question:

How does revenue change across months and years?

Purpose:

Identifies seasonality and long-term sales trends.

```
SELECT
    year,
    month,
    SUM(total) AS revenue
FROM walmart
GROUP BY year, month
ORDER BY year, month;
```

Output:

	year integer	month integer	revenue double precision
1	2019	1	82440.540000000008
2	2019	2	60161.659999999998
3	2019	3	69285
4	2019	4	7570.6000000000005

Insight:

Revenue fluctuates across months, indicating **seasonality rather than steady growth**.

Business Implication:

Seasonal revenue patterns should guide forecasting, inventory planning, and promotional calendars.

5. ADVANCED KPIs & STRATEGIC INSIGHTS

Objective: Support decision-making and risk identification.

5.1 Busiest day for each branch

Business Question:

Which day has the highest traffic at each branch?

Purpose:

Enables branch-level demand planning.

```
WITH daily_rank AS (  
    SELECT  
        branch,  
        weekday AS day_of_week,  
        COUNT(*) AS num_transactions,  
        RANK() OVER (PARTITION BY branch ORDER BY COUNT(*) DESC)  
    AS rank  
    FROM walmart  
    GROUP BY branch, weekday  
)  
SELECT * FROM daily_rank WHERE rank = 1;
```

Output:

	branch text	day_of_week text	num_transactions bigint	rank bigint
1	WALM0...	Saturday	14	1
2	WALM0...	Thursday	14	1
3	WALM0...	Sunday	15	1
4	WALM0...	Sunday	20	1

Insight:

Each branch has a **different busiest weekday**, indicating localized demand patterns.

Business Implication:

Branch-specific demand patterns require customized staffing and operational planning.

5.2 Branches with revenue decline (2022 → 2023)

Business Question:

Which branches show year-over-year revenue decline?

Purpose:

Flags underperforming branches for deeper investigation.

```
WITH rev_2022 AS (  
    SELECT branch, SUM(total) AS revenue_2022  
    FROM walmart  
    WHERE year = 2022  
    GROUP BY branch  
) ,  
rev_2023 AS (  
    SELECT branch, SUM(total) AS revenue_2023  
    FROM walmart  
    WHERE year = 2023  
    GROUP BY branch  
)  
SELECT  
    rev_2022.branch,  
    rev_2022.revenue_2022,  
    rev_2023.revenue_2023,  
    ROUND(  
        ((rev_2022.revenue_2022 - rev_2023.revenue_2023) /  
rev_2022.revenue_2022)::numeric,  
        2  
    ) * 100 AS decline_pct  
FROM rev_2022  
JOIN rev_2023 ON rev_2022.branch = rev_2023.branch  
WHERE rev_2023.revenue_2023 < rev_2022.revenue_2022  
ORDER BY decline_pct DESC  
LIMIT 5;
```

Output:

	branch text	revenue_2022 double precision	revenue_2023 double precision	decline_pct numeric
1	WALM0...	1731	647	63.00
2	WALM0...	2581	1069	59.00
3	WALM0...	2446	1030	58.00
4	WALM0...	2099	931	56.00
5	WALM0...	1723	850	51.00

Insight:

Some branches experienced **50–63% year-over-year revenue decline**, signaling serious underperformance.

Business Implication:

Declining branches should be prioritized for performance review and corrective action to prevent long-term revenue loss.

5.3 Average basket value

Business Question:

What is the average transaction value?

Purpose:

Measures customer spending behavior.

```
SELECT
    AVG(total) AS avg_basket_value
FROM walmart;
```

Output:

	avg_basket_value double precision
1	121.34881933995389

Insight:

Customers spend **~121 per transaction on average**, indicating moderate basket sizes.

Business Implication:

There is potential to increase revenue by encouraging upselling and bundled purchases.

5.4 Customer segmentation by purchase value

Business Question:

How are transactions distributed by spend level?

Purpose:

Supports targeted pricing and promotional strategies.

```
SELECT
  CASE
    WHEN total < 100 THEN 'Low'
    WHEN total BETWEEN 100 AND 300 THEN 'Medium'
    ELSE 'High'
  END AS customer_segment,
  COUNT(*) AS num_customers,
  SUM(total) AS revenue
FROM walmart
GROUP BY customer_segment
ORDER BY revenue DESC;
```

Output:

	customer_segment text	num_customers bigint	revenue double precision
1	Medium	4118	676427.87
2	Low	5440	310678.91000000003
3	High	411	222619.59999999999

Insight:

Medium-value customers drive most revenue, not high spenders.

Business Implication:

Marketing strategies should focus on medium-value customers who contribute the highest share of revenue.

5.5 Most common invoice value ranges

Business Question:

What transaction value ranges occur most frequently?

Purpose:

Helps design bundles and pricing thresholds.

```
SELECT
    ROUND(total::numeric, -1) AS rounded_total,
    COUNT(*) AS freq
FROM walmart
GROUP BY rounded_total
ORDER BY freq DESC
LIMIT 5;
```

Output:

	rounded_total numeric	freq bigint
1	50	821
2	70	807
3	80	787
4	60	735
5	40	668

Insight:

Most transactions cluster around **\$40–\$80**, with \$50 being the most frequent.

Business Implication:

Pricing and promotional thresholds should be designed around common spending ranges to drive incremental sales.

5.6 City-wise average profit margin

Business Question:

Which cities are more profitable on average?

Purpose:

Supports geographic performance comparison and expansion strategy.

```
SELECT
    city,
    ROUND((AVG(profit_margin) * 100)::numeric, 2) AS
avg_profit_percent
FROM walmart
GROUP BY city
ORDER BY avg_profit_percent DESC;
```

Output:

	city text	avg_profit_percent numeric
1	Mansfield	57.00
2	New Braunfels	51.57
3	Frisco	48.00
4	Amarillo	48.00
5	Irving	46.00

Insight:

Profitability varies significantly by city, with **Mansfield (~57%)** and **New Braunfels (~51%)** showing the highest average profit margins, while other cities perform notably lower.

Business Implication:

Investment and expansion decisions should prioritize cities with consistently higher profit margins.
